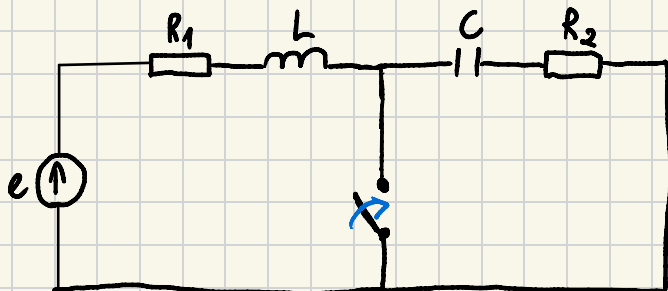


ĆWICZENIA 12

METODA KLASYCZNA

Zad. 1



stan ustalony
po przełączeniu

$$u_C(t) = u_{Cu}(t) + u_{Cp}(t)$$

↑
stan przejściowy

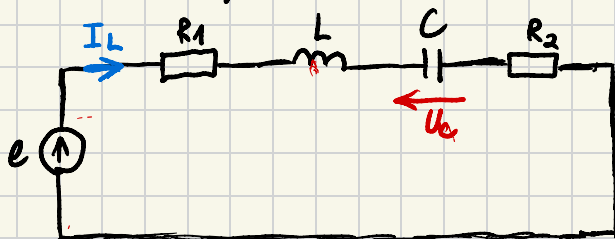
$$e(t) = 10\sqrt{2} \sin(10t + 90^\circ)$$

$$L = 2 \text{ H} \quad C = 10 \text{ nF} = 10^{-2} \text{ F}$$

$$R_1 = R_2 = 5 \Omega$$

Warunki początkowe to normalna analiza obwodu, gdyż znamy

1° Warunki początkowe (Układ przed przełączeniem) A.D. 1



$$E = 10e^{j90^\circ} \Rightarrow \omega = 10$$

$$Z_L = j\omega L = j20$$

$$Z_C = -j\frac{1}{\omega C} = -j \cdot \frac{1}{10 \cdot 10^{-2}} = -j10$$

$$R_1 = R_2 = 5 \Omega$$

$$Z_2 = R_1 + Z_L + Z_C + R_2 =$$

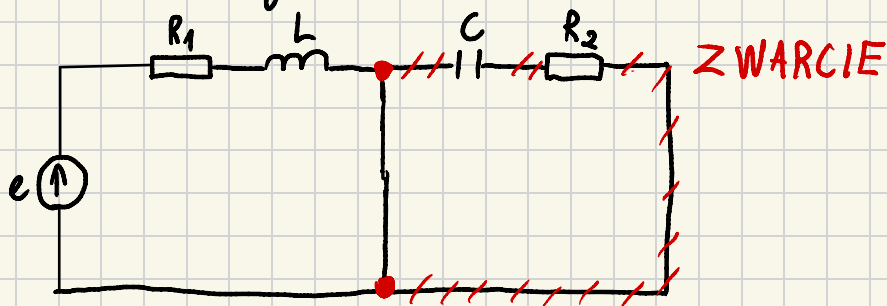
$$= 5 + j20 - j10 + 5 = 10 + j10 = 10\sqrt{2}e^{j45^\circ}$$

$$I_L = \frac{E}{Z_2} = \frac{10e^{j90^\circ}}{10\sqrt{2}e^{j45^\circ}} = \frac{\sqrt{2}}{2}e^{j45^\circ} \Rightarrow i_L(t) = 1 \sin(10t + 45^\circ)$$

$$U_C = I_L \cdot Z_C = \frac{\sqrt{2}}{2}e^{j45^\circ} \cdot (-j10) = 5\sqrt{2}e^{-j45^\circ} \Rightarrow u_C(t) = 10 \sin(10t - 45^\circ)$$

$$i_L(0^-) = 1 \cdot \sin 45^\circ = \frac{\sqrt{2}}{2} \text{ [A]} \quad u_C(0^-) = 10 \sin(-45^\circ) = -5\sqrt{2} \text{ [V]}$$

2° Stan ustalony po przełączeniu



$u_{Ck}(t) = 0 [V]$ ← Na gałęzi z kondensatorem jest zwarcie

$u_{Ck}(0^+) = 0 [V]$

$Z_z = R_1 + Z_L = 5 + j20 [\Omega]$

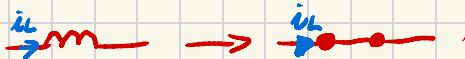
$I_L = \frac{E}{Z_z} = \frac{10e^{j30^\circ}}{5 + j20} = 0,485e^{j14^\circ} = 0,49e^{j14^\circ}$

$i_{Lk}(t) = 0,49 \cdot \sqrt{2} \sin(10t + 14^\circ) = 0,693 \sin(10t + 14^\circ)$

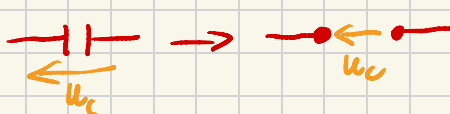
$i_{Lk}(0^+) = 0,693 \sin 14^\circ = 0,168 [A]$

AD 1. Jeśli $e(t) \sim \text{const.}$, to wtedy pamiętamy, że w punktach 1° i 2°:

- cewka stanowi zwarcie



- kondensator stanowi przerwę



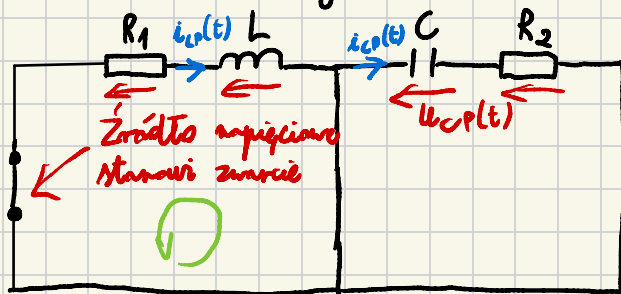
Uzasadnienie „matematyczne”

$e(t) \sim \text{const.} \Rightarrow \omega = 0$

$Z_L = j\omega L = j \cdot 0 \cdot L = 0 \rightarrow \text{zwarcie}$

$Z_C = -j \frac{1}{\omega C} = -j \cdot \frac{1}{0 \cdot C} = \frac{1}{0} = \infty$ przerwa

3° Składana przejściowa



3.1° Wyznaczam $i_{LP}(t)$

$$U_{R1} + U_L = 0$$

$$R_1 \cdot i_{LP}(t) + L \cdot \frac{di_{LP}(t)}{dt} = 0$$

$$R_1 \cdot 1 + L \cdot \lambda = 0$$

$$\lambda = -\frac{R_1}{L} = -\frac{5}{2} = -2,5$$

Podkreślane na zielono w 2°

$$i_{LP}(t) = A_1 e^{\lambda t}$$

$$i_{LP}(t) = A_1 e^{-2,5t}$$

WAŻNE!!!

$$i_L(t) = i_{Lw}(t) + i_{LP}(t)$$

$$i_L(0^-) = i_{Lw}(0^+) + i_{LP}(0^+)$$

↑
warunek początkowy

↑
stan ustalony po przekroczeniu

$$\frac{\sqrt{2}}{2} = 0,168 + A_1 e^{-2,5 \cdot 0}$$

$$A_1 = \frac{\sqrt{2}}{2} - 0,168 = 0,54$$

$$i_{LP}(t) = 0,54 e^{-2,5t} [A]$$

$$i_L(t) = i_{Lw}(t) + i_{LP}(t) = 0,693 \sin(10t + 14^\circ) + 0,54 e^{-2,5t}$$

3.2.^o Wyznaczam $u_{cp}(t)$

$$U_c + U_{R2} = 0$$

$$u_{cp}(t) + R_2 \cdot i_{cp}(t) = 0$$

$$u_{cp}(t) + R_2 \cdot C \cdot \frac{du_{cp}(t)}{dt} = 0$$

$$1 + R_2 \cdot C \cdot s = 0$$

$$s = -\frac{1}{R_2 C} = -\frac{1}{5 \cdot 10^{-2}} = -20$$

$$u_{cp}(t) = A_2 e^{st} = A_2 e^{-20t}$$

Ta sama logika, jak wcześniej

$$u_c(0^-) = u_{cu}(0^+) + u_{cp}(0^+)$$

$$-5\sqrt{2} = 0 + A_2$$

$$A_2 = -5\sqrt{2}$$

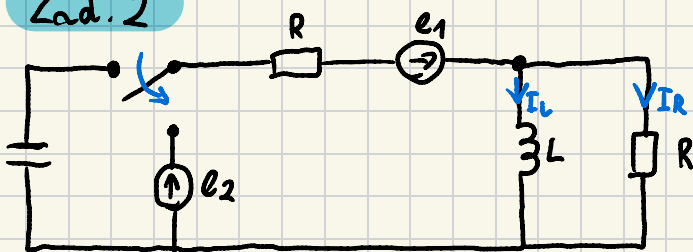
$$u_{cp}(t) = -5\sqrt{2} e^{-20t} [V]$$

$$u_c(t) = u_{cu}(t) + u_{cp}(t) = -5\sqrt{2} e^{-20t}$$

ODP: $i_c(t) = 0,693 \sin(10t + 14^\circ) + 0,54 e^{-25t}$

$$u_c(t) = -5\sqrt{2} e^{-20t}$$

Zad. 2



$$e_1(t) = 10\sqrt{2} \sin(t - 90^\circ) \text{ [V]}$$

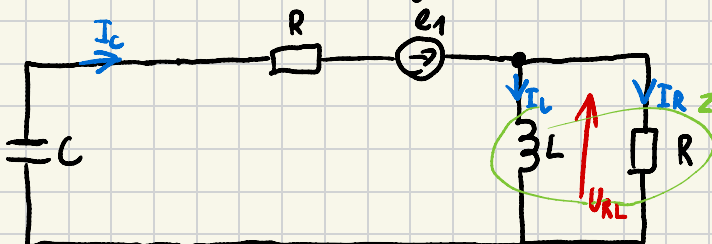
$$e_2(t) = 5 \text{ [V]}$$

$$R = 5 \text{ [\Omega]}$$

$$L = 1 \text{ [H]}$$

$$C = 0,5 \text{ [F]}$$

1° Warunek porażki



$$Z_{RL} = \frac{R \cdot Z_L}{R + Z_L} = \frac{5 \cdot j}{5 + j} = \frac{5}{26} + j\frac{5}{26}$$

$$\omega_1 = 1 \text{ [rad/s]}$$

$$Z_L = j\omega_1 L = j1$$

$$Z_C = -j\frac{1}{\omega_1 C} = -j\frac{1}{1 \cdot 0,5} = -j2$$

$$E = 10e^{-j90^\circ} = -j10$$

$$Z_Z = Z_C + Z_R + Z_{RL} = -j2 + 5 + \frac{5}{26} + j\frac{5}{26} = 5,19 - j1,808$$

$$I_C = \frac{E_1}{Z_Z} = \frac{-j10}{5,19 - j1,808} = 1,82 e^{-j71^\circ}$$

$$U_C = Z_C \cdot I_C = -j2 \cdot 1,82 e^{-j71^\circ} =$$

RC

$i_C(t)$

$$u_{Cp}(t) + R_2 \cdot C \frac{du_{Cp}(t)}{dt} = 0$$

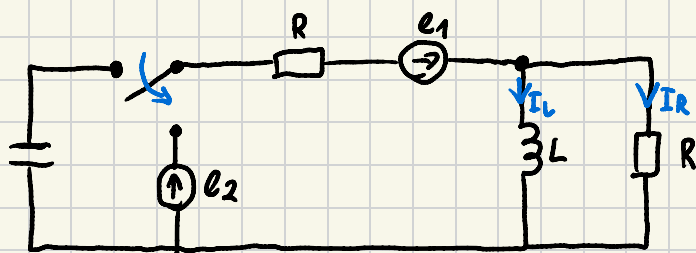
$$1 + 5 \cdot 0,01 \lambda = 0 \Rightarrow \lambda = -20$$

$$u_{Cp}(t) = A_2 e^{-20t}$$

$$-5\sqrt{2} = 0 + A_2 e^0 \Rightarrow A_2 = -5\sqrt{2}$$

$$u_C(t) = \underbrace{0}_{u_{CC}(t)} - \underbrace{5\sqrt{2}}_{u_{Cp}(t)} e^{-20t}$$

Zad. 2 DO ZROBIENIA !!!



$$i_L(t) = ?$$

$$e_1(t) = 10\sqrt{2} \sin(t - 90^\circ) \text{ [V]} \quad 1^\circ \text{ Warunki początkowe}$$

$$e_2(t) = 5 \text{ [V]} \quad I = 0,37 - j1,85$$

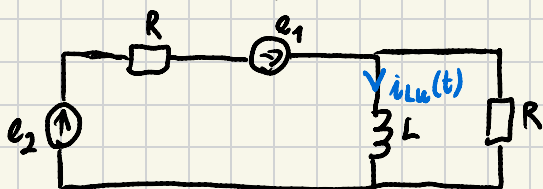
$$R = 5 \text{ [\Omega]} \quad I_L = -j1,85185$$

$$L = 1 \text{ [H]} \quad I_R = 0,37037$$

$$C = 0,5 \text{ [F]} \quad i_L(t) = 1,85185 \cdot \sqrt{2} \sin(t - 90^\circ)$$

$$i_L(0^-) = -2,6189 \text{ [A]}$$

2° Stan ustalony pracy obwodu po przetworzeniu



UWAGA!

Nie może $e_1 + e_2 = 0$ (patrzyć źródła)
bo inna pulsacja ($\omega_1 \neq \omega_2$)

a) Dla e_1

$$I_{Lu}' = -1,6896 - j1,72414$$

$$i_{Lu}'(t) = 1,8569\sqrt{2} \sin(t - 111,8^\circ) [A]$$

b) Dla e_2

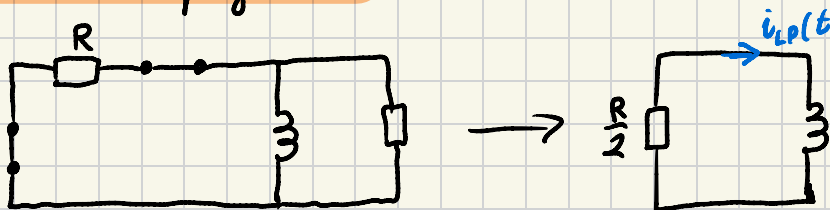
$$I_{Lu}'' = \frac{E_2}{R} = 1 [A]$$

$$i_{Lu}''(t) = 1 [A]$$

$$i_{Lu}(t) = 1 + 1,8569\sqrt{2} \sin(t - 111,8^\circ)$$

$$i_{Lu}(0^+) = 1 + (-2,4383) = -1,4383 [A]$$

3° Składnik przejściowy



$$\frac{R}{2} \cdot i_{LP}(t) + L \frac{di_{LP}(t)}{dt} = 0$$

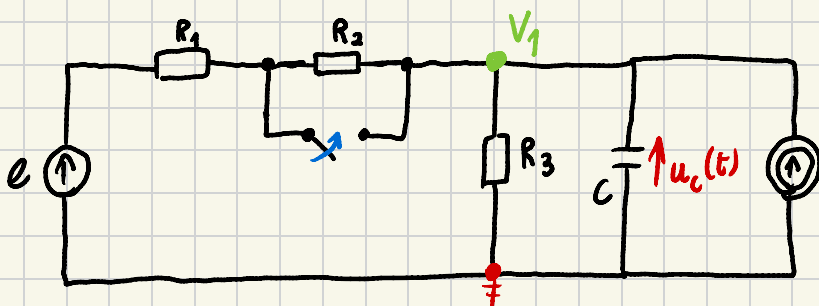
$$-2,6189 = -1,4383 + A \Rightarrow A = -1,18$$

$$2,5 + 1 \cdot s = 0 \Rightarrow s = -2,5$$

$$i_{LP}(t) = A e^{-2,5t}$$

$$i_L(t) = 1 + 1,8569\sqrt{2} \sin(t + 111,8^\circ) - 1,18 e^{-2,5t}$$

Zad. 3 DO ZROBIENIA !!!



$$E = 120 \text{ V} \quad R_1 = R_2 = 6 \, \Omega$$

$$J = 4 \text{ A} \quad R_3 = 12 \, \Omega$$

$$C = 5 \, \mu\text{F}$$

1° Warunki początkowe

$$V_1 \left(\frac{1}{R_3} + \frac{1}{R_1 + R_2} \right) = \frac{E}{R_1 + R_2} + J$$

$$V_1 = 84 \text{ [V]}$$

$$u_c(t) = 84 \text{ [V]}$$

$$u_c(0^-) = 84 \text{ [V]}$$

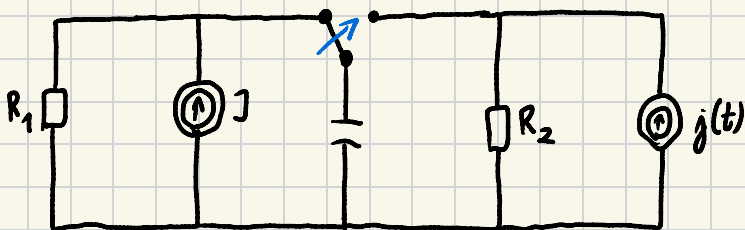
$$u_c(t) =$$

2° Stan ustalony po przełączeniu

$$V_1 \left(\frac{1}{R_1} + \frac{1}{R_3} \right) = \frac{E}{R_1} + J$$

$$V_1 = 96 \text{ [V]}$$

Zad. 4



GDY ŹRÓDŁO
PRĄDOWE STAŁE
TO KONDENSATOR
JEST PRZERWĄ

$$j(t) = 20 \sin(t + 90^\circ)$$

$$R_1 = R_2 = 10 \Omega$$

$$J = 4 \text{ A} \quad C = 100 \text{ nF}$$

2° Stan ustalony po połączeniu

1° Warunki początkowe

$$u_C = R_i J = 10 \cdot 4 = 40 \text{ [V]}$$

$$u_C(t) = 40 \text{ [V]}$$

$$u_C(0^-) = 40 \text{ [V]}$$